MILITARY SPECIFICATION

TEST FIXTURE SETS, ELECTRONIC

1. SCOPE

1.1 This exhibit covers requirements for design and fabrication of electronic test fixture sets. A test fixture set is a complete set, specifically designed for testing significant electrical characteristics of module, parts or subassemblies when such characteristics cannot be tested practically by using standard or common test equipment. (See 6.1).

1.2 A complete test fixture set comprises the following items:

(a) A separate test fixture (which must be packaged by itself) for each module, subassembly, etc. cited in the contract under the test fixture requirement.

(b) Power supply for each test fixture, if not integral with the test fixture.

(c) Special signal generator, if a standard commercial or standard military signal generator cannot be used with the test fixture.

(d) Special output indicating instrument, if not integral with the test fixture or if a standard commercial or standard military instrument cannot be used.

(e) Control standard (nominal), two each.

(f) Maintenance parts.

(g) Set of special tools, and other special devices such as connecting cables for each commercial test equipment required.

(h) Required cabinets to contain the foregoing, except as otherwise specified in 3.6 and 3.9.2.

FSC 6625

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this exhibit to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-11268	Parts, Materials, and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items.

STANDARDS

MILITARY

MIL-STD-280	Definition of Terms for Equipment Divisions.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Definitions.- Standard MIL-STD-280 applies to terms for equipment divisions such as part, subassembly, assembly, and unit. In addition, the following special definition shall apply:

3.1.1 Control Standards.- A control standard is a device simulating the part of subassembly to be tested by the test fixture set and embodying the nominal value of a specific performance characteristic of the part or subassembly

3.2 General.- The test fixture set shall provide suitable and convenient means for testing selected electrical characteristics of modules, parts or subassemblies on production runs, to assure interchangeability for those characteristics when the modules, parts or subassemblies are employed in equipment units.

3.2.1 Parts, materials, processes, and construction.- Except as otherwise necessary in order to meet the requirements of this exhibit, the parts, materials, and processes used in fabrication of the test fixture set shall conform to Specification MIL-P-11268 and shall be specifically selected to promote performance stability of the set. Construction of the test fixture set shall be such as to provide the greatest practicable stability of performance.

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3.2.2 Electrical power.- When practicable, the power supply shall be integral with the test fixture; otherwise, the power supply shall be a separate unit of the test fixture set. The test fixture set shall operate from nominal 115-volt, 50/60-cps line, and shall provide all power required for operation of the test fixture. Each output voltage of the power supply shall be metered and provided with a means for adjustment to its nominal value. Input to the power supply shall be protected by fuses or circuit breakers connected directly to all input terminals. Where practicable, other circuits shall be protected by the same means to prevent damage from incorrect adjustment, short or open circuits, or failure of tubes or other parts. With the test fixture set connected for operation, personnel shall be protected from contact with potentials in excess of 30 volts to ground, chassis, or frame, including potentials on charged capacitors.

3.2.3 Marking.- Except as otherwise specified herein, the various items of the test fixture set shall be marked in accordance with Specification MIL-M-13231. In addition, each item except the test fixture shall be marked "USED WITH ELECTRONIC TEST FIXTURE NR. _____."

3.3 Test Fixture.- The test fixture shall simulate, as nearly as possible, that portion of equipment unit which employs the module, part or subassembly to be tested.

3.3.1 Electrical circuitry and characteristics of the test fixture shall duplicate as nearly as possible that portion of the equipment unit which is concerned with the module, part or subassembly, as necessary to measure the electrical characteristics selected for test and assure compliance with the end item specification. Parts and subassemblies used in construction of the test fixture shall be identical to corresponding items in the equipment unit which it simulates, except as necessary to make a substitution in order to improve stability of the test fixture. In such instance, approval of the contracting officer shall be requested in writing, giving necessary details regarding the substitute item and why the substitution is desirable.

3.3.2 The method used to connect the part or subassembly to be tested into the test fixture shall be convenient in use and shall insure firm electrical connection and mechanical support while the part or subassembly is being tested. Connectors used for this purpose shall be at least equal in quality to the corresponding connectors used in the equipment on contract.

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3.3.3 In addition to other specified marking, the test fixture shall be marked as follows:

"FOR TESTING THE (__A__) OF (__B__),
WHICH IS USED IN (__C__)"

The blank spaces shown above shall be treated as follows:

A. Insert kind of test, such as IMPEDANCE.
B. Insert name of module, part or subassembly to be tested.
C. Insert Government type number of the assembly or unit in which the module, part or subassembly is used.

3.4 Signal Generator.- When practicable, provision shall be made to permit use of a standard commercial or standard military signal generator for injecting the required signals into the test fixture; and suitable connectors shall be provided for the purpose. When a standard commercial signal generator cannot be used, the contractor may obtain information regarding availability of a suitable standard military signal generator, from the Contracting Officers Technical Representative. When use of a standard commercial or standard military signal generator is not practicable, the test fixture set shall include a signal generator as a separate unit or integral with the test fixture.

3.5 Output indicating instrument.- When practicable, provision shall be made to permit use of a standard commercial or standard military indicating instrument for measuring the output of the test fixture; and suitable connectors shall be provided for that purpose. When a standard commercial indicating instrument cannot be used, the contractor may obtain information regarding availability of a suitable standard military indicating instrument, from the Contracting Officers Technical Representative. When use of a standard commercial or standard military indicating instrument is not practicable, the test fixture set shall include an indicating instrument as a separate unit or integral with the test fixture.

3.6 Control standards.- Nominal control standards shall be furnished as needed for characteristic to be measured. The control standard shall duplicate the circuitry and functions of the module, part or subassembly which it simulates; however, substitute materials or parts may be used, or parts may be added or specially selected, to improve stability of the control standard or obtain the desired value of the characteristic to be tested by the fixture. In addition to other specified marking, each control standard shall be marked "CONTROL STANDARD FOR MEASURING (electrical characteristic)." One standard is to be stored in the test fixture and the remaining standard is to be shipped to place designated by the Contracting Officers Technical Representative.

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3.7 Maintenance parts.- The Government may require maintenance parts, in which case a list of such parts will be furnished by the contracting officer, ie, tubes, fuses, etc.

3.8 Special tools and devices.- Special tools and other special devices needed for calibration, operation, and maintenance of the test fixture set shall be an item thereof.

3.9 Enclosures.-

3.9.1 Cabinets.- Test fixtures, separate power supplies and other items requiring protection from environmental and physical damage shall be enclosed in suitable, individual metal cabinets and shall be so designed to contain and protect the items enclosed, ie; standard, connecting cables, test procedure, spare parts, etc., of the test fixture set during trucking and manual handling around a manufacturing plant and during transportation and storage. Cases shall retain their contents firmly in place when the lids are closed, case lids shall be held closed by means of catches of the drawn pull type.

3.9.2 If practicable, a single cabinet shall be provided for the entire test fixture set. However, no single cabinet with contents shall weigh more than 40 pounds, unless unavoidable and unless approval to exceed the 40-pound limit is obtained from the Contracting Officers Technical Representative.

3.9.3 In addition to other specified marking, the following shall appear on each case:

(a) "CAUTION—ELECTRONIC EQUIPMENT—HANDLE WITH CARE REPLACE CONTENTS IN ASSIGNED POSITION IN CASE, WHEN NOT IN ACTUAL USE." This caution notice shall be placed in a prominent location on the outside of the case.

(b) "CASE 1 OF 3". "CASE 2 OF 3", etc, as applicable.

3.10 Design and drawings of test fixture sets.-

3.10.1 Design.- The contractor shall be responsible for design of the test fixture sets, unless the contract stipulates that drawings thereof will be furnished by the Government.

3.11 Correlation data.-

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3.11.1 The contractor is to perform the necessary measurements and certified correlation data shall be submitted along with the test fixture set. This correlation data shall show that modules, parts or subassemblies which have been tested in and found satisfactory by the test fixture set will perform satisfactorily in the unit in which they are to be used.

4. TESTS

4.1 Standard Conditions.- Except as otherwise specified, calibration and tests shall be conducted under the following standard conditions:

(a) Temperature: Factory Ambient.

(b) Relative humidity: Factory Ambient.

(c) Warmup: Warmup period of 15 minutes with all supply voltages at their nominal values.

(d) Supply voltages: Nominal values cited in the Test Procedures.

4.1.1 Controlled voltages.- Each controlled voltage shall be tested to determine that it can be adjusted to its nominal value.

4.1.2 Two-hour operation.- The test fixture shall be calibrated and then operated for 2 hours. Calibration of the test fixture shall have changed not more than ±0.5 percent as a result of this test.

5. PREPARATION FOR DELIVERY

5.1 Each submission and delivery shall be packaged and packed in accordance with best commercial practice, and in a manner that will afford protection against corrosion, deterioration, and physical damage during direct shipment to the first receiving activity.

6. NOTES

6.1 Intended use.- The test fixture sets, electronic, covered by specification are intended for use by Government inspection of ECOM material.

CUSTODIAN
ARMY-EL

PREPARING ACTIVITY
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PROJECT NO. 6625-A230

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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